MILITARY SPECIFICATION

CASE, CARTRIDGE, 105MM, M115B1

This amendment forms a part of Military Specification MIL-C-63296 (AR) dated 22 January 1980 and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: STANDARDS MILITARY

Add "MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes".

PAGE 4

3.8.4:

Delete "No cartridge case shall have a burr." and substitute "Burrs shall be limited to those described in 105mm Cartridge Case 'Visual Defect Standards'."

PAGE 5

4.3.1:

Change "254*" to "262".

Delete the following sentence in its entirety: "*All samples must meet the requirements for thermal treatment and high stress expansion test as performed in the uncoated state."

AMSC N/A
FSC 1315
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PAGE 6

* Table I - First Article Inspection:

Cartridge Case - Prior to Coating: Delete in its entirety.

Cartridge Case - After Coating:

Change "254" to "262"

Add:	"Tension Test (Info Only)	4	3.3	4.5.4
	Macro Examination	4	3.8.2	4.5.5
	Hardness	20	3.3	4.5.3"

PAGE 8

* 4.4.2.1: Delete in its entirety and substitute new paragraph 4.4.2.1 (included herein as Page 3).

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* 4.4.2.2: Delete in its entirety and substitute new paragraph 4.4.2.2 (included herein as Pages 4 through 6).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10520214
4.4.2.1	Case, Cartridge - Prior to Coating				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	High stress expansion test		100%(a)	3.1	4.5.2

NOTES:
(a) All non-compliance units must be removed from the lot. Evidence of cracks in any portion of the case shall be cause for rejection.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating		10520214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Cracks		100%(a)	3.8.1	4.5.7
102	Metal defective		100%(a)	3.8.1	4.5.7
103	Thickness of coating	(See 4.5.6)		3.4	4.5.6.1
104	Salt spray	(See 4.5.6)		3.1	4.5.6.2
105	Maximum profile and alignment		100%(a)	3.5	4.5.8
106	Excess pressure ballistic test	(See 4.4.3.1.2)		3.6	4.5.9.2.1
107	True position of pitch diameter of thread with body		0.40%	3.1	Gage
108	True position of primer hole counterbore with pitch diameter of thread		0.40%	3.1	Gage
109	True position of primer hole thread relief with pitch diameter of thread		0.40%	3.1	Gage
110	Pitch diameter of thread, max.		0.40%	3.1	Gage
111	Minor diameter of thread, max.		0.40%	3.1	Gage
112	Length total, max.		0.40%	3.1	Gage
113	Diameter of primer hole counterbore		0.40%	3.1	Gage
114	Depth of primer hole counterbore, min.		0.40%	3.1	Gage
115	Length from flange to bottom of primer hole counterbore, max.		0.40%	3.1	Gage
116	Depth of thread relief		0.40%	3.1	Gage
117	Protective coating defective		0.40%	3.4	Visual
118	Thermal treatment		100%(a)	3.3.1	4.5.1

NOTES:
(a) All non-compliance units must be removed from the lot.

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 10520214
4.4.2.2	Case, Cartridge - After Coating				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
119	Hardness	4	(b)	3.3	4.5.3
120	Macro-Examination	4	(b)	3.8.2	4.5.5
121	Diameter of flange, min.		0.40%	3.1	Gage
122	Diameter of mouth		0.40%	3.1	Gage
123	Length of mouth diameter, min.		0.40%	3.1	Gage
124	Inside diameter before taper at mouth end		0.40%	3.1	Gage
125	True position of inside diameter before taper at mouth end with mouth diameter		0.40%	3.1	Gage
126	Thickness of flange, min.		0.40%	3.1	Gage
127	Thickness of base, min.		0.40%	3.1	Gage
128	Wall thickness at first datum from base		0.40%	3.1	Gage
129	Wall thickness at second datum from base		0.40%	3.1	Gage
130	Wall thickness at third datum from base		0.40%	3.1	Gage
131	Wall thickness at mouth		0.40%	3.1	Gage
132	Contour of inside of base and sidewall not smooth		0.40%	3.1	Visual
MINOR					
201	Pitch diameter of thread, min.		0.65%	3.1	Gage
202	Diameter of primer hole thread relief		0.65%	3.1	Gage
203	Thread damaged or not full		0.65%	3.8.3	Visual
204	Body dented or damaged		0.65%	3.1	Visual
205	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:
(b) Reject on one defect.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating		10520214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
206	Wax coating missing		0.65%	3.1	Visual
207	Evidence of poor workmanship		0.65%	3.8.4 & 3.8.5	Visual
208	Diameter of body under flange, min.		0.65%	3.1	Gage
209	Location of first basic diameter from flange, min.		0.65%	3.1	Gage
210	Location of second basic diameter from flange, min.		0.65%	3.1	Gage
211	Location of third basic diameter from flange, min.		0.65%	3.1	Gage
212	Location of fourth basic diameter from flange, min.		0.65%	3.1	Gage
213	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
214	Surface finish improper		0.65%	3.1	Visual
215	Length total, min.		0.65%	3.1	Gage
SPECIAL					
301	Tension Test (Information Only)	(c)		3.3	4.5.4

NOTES: (c) Samples used for the hardness test shall be used in the tension test (see 4.5.4).

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4.4.2.3:

Major Defect, delete "None defined."

Add Major Defect 101:

"101	Cracks and other metal defects (cartridge cases) just prior to packing into carton	100%	3.8.1	Visual"

PAGE 15

4.5.4: Delete this paragraph in its entirety and substitute:

"Tension test (Information Only). Four cartridge cases from each lot shall be subjected to this test. The test specimens shall be taken from each cartridge case in accordance with the standard position and size of the tensile specimen chart described in Figure 1. The yield strength, tensile strength and elongation shall be determined in accordance with the procedures prescribed by ASTM-E8. The results shall be recorded for information only."

4.5.6.1:

Last sentence, Delete "...rejection of the lot." and substitute "...testing additional samples in accordance with MIL-STD-105, Inspection Level II, 0.40% AQL."

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